EXAM PREPARATION · FALL 2026

Intermediate Macroeconomics

Every exercise from Tutorials 2, 4 and 5 — consumption, unemployment and real-business-cycle theory — with formula catalogs, complete step-by-step derivations and all calculations. Compiled for exam preparation.

Question bank · Formula catalog · Full solutions

Sources: Tutorial 2 (Consumption) · Tutorial 4 (Unemployment) · Tutorial 5 (Real-Business-Cycle Theory) · Lecture 5 — Efficiency Wages

Contents

1 How to Use This Document **2**

2 Theory Essentials — Formula Catalog **2**

2.1 Two-period consumption (Fisher) . . . 2

2.2 Life-cycle saving (Modigliani–Brumberg) and permanent income . . . 3

2.3 Efficiency wages (Lecture 5) . . . 3

2.4 Real business cycles (Tutorial 5) . . . 4

3 Tutorial 2 — Consumption: Complete Solutions **5**

3.1 Fisher's model with different saving and borrowing rates . . . 5

3.2 Log utility: Lagrangian, closed-form solution, and a numerical example . . . 6

3.3 Life-cycle saving (Modigliani and Brumberg, 1954) . . . 7

4 Tutorial 4 — Unemployment: Complete Solutions **9**

4.1 Shapiro–Stiglitz value functions . . . 9

4.2 Comparative statics in the Shapiro–Stiglitz model . . . 9

4.3 Union wage premiums and efficiency wages (Summers, 1988) . . . 11

4.4 Wage rigidity in the efficiency-wage model (Lecture 5, eqs. 18–19) . . . 12

5 Tutorial 5 — Real-Business-Cycle Theory: Complete Solutions **12**

5.1 Technology shocks and macroeconomic fluctuations in the RBC model . . . 12

5.2 A simplified RBC model with additive technology shocks (Blanchard–Fischer) . . . 14

6 Exam Crib Sheet — All Key Results on One Page **17**

1 How to Use This Document

This guide prepares you for the **Intermediate Macroeconomics** exam. It covers every exercise from **Tutorial 2 (Consumption)**, **Tutorial 4 (Unemployment)** and **Tutorial 5 (Real-Business-Cycle Theory)**, plus the efficiency-wage model of **Lecture 5**. Every problem is reproduced with its original numbers and solved step by step: setup, key formulas, full derivation, final result.

Source	Models	Key results
Tutorial 2	Fisher two-period model with $r_s \neq r_b$; Lagrangian with ln utility; life-cycle saving (Modigliani)	$C_1 = \frac{1}{1+\beta}\left(Y_1 + \frac{Y_2}{1+r}\right)$; borrower/saver geometry; $W(R) = (T-R)C$
Tutorial 4	Shapiro–Stiglitz value functions; NSC comparative statics; union wage premium (Summers)	$\rho V_S = w - (b+q)(V_S - V_U)$; $u = \frac{\beta+\mu f}{b(1+\mu f)} = 8.1\%$
Tutorial 5	RBC impulse responses; log-linearized policy rules; simplified RBC (Blanchard–Fischer)	$\tilde{A}_t = 0.01(0.95)^t$; $E[C_{t+1}] = C_t$ (random walk); $\gamma = \frac{A}{1-\phi+A}$
Lecture 5	Efficiency wages: Solow condition, example $e = \left(\frac{w-x}{x}\right)^{\beta}$, wage rigidity	$we'(w)/e(w) = 1$; $w^* = \frac{x}{1-\beta}$; $u_{EQ} = \beta/b$

Notation. $\hat{x}_t \equiv \ln x_t - \ln \bar{x}_t$ denotes the log deviation of a variable from its balanced-growth-path (BGP) value; $E_t[\cdot]$ is the expectation conditional on period-t information; a prime on a function (e') is a derivative, while C_{t+1} is a time subscript.

2 Theory Essentials — Formula Catalog

2.1 Two-period consumption (Fisher)

Budget constraints. With saving rate r_s and borrowing rate r_b, $r_s < r_b$, the consumer faces a kinked constraint. If he consumes less than income in period 1 (saver):

$$C_1 + \frac{C_2}{1+r_s} = Y_1 + \frac{Y_2}{1+r_s}. \tag{1}$$

If he consumes more than income in period 1 (borrower):

$$C_1 + \frac{C_2}{1+r_b} = Y_1 + \frac{Y_2}{1+r_b}. \tag{2}$$

Both lines pass through the endowment point (Y_1, Y_2); the feasible set is the area between them. First-period consumption depends on *present value at the relevant rate*: $Y_1 + \frac{Y_2}{1+r_s}$ for a saver, $Y_1 + \frac{Y_2}{1+r_b}$ for a borrower, and only Y_1 for a consumer at the kink.

Optimal choice. Maximize $U(C_1, C_2) = \ln C_1 + \beta \ln C_2$, $0 < \beta < 1$, subject to $C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$. The Lagrangian and first-order conditions:

$$\mathcal{L} = \ln C_1 + \beta \ln C_2 + \lambda\left(Y_1 + \tfrac{Y_2}{1+r} - C_1 - \tfrac{C_2}{1+r}\right), \tag{3}$$

$$\frac{\partial \mathcal{L}}{\partial C_1} = \frac{1}{C_1} - \lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial C_2} = \frac{\beta}{C_2} - \frac{\lambda}{1+r} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = 0. \tag{4}$$

Eliminating λ gives the **Euler equation** (intertemporal optimality):

$$\frac{1}{C_1} = \frac{\beta(1+r)}{C_2} \qquad \Longrightarrow \qquad \boxed{C_2 = (1+r)\,\beta\, C_1} \tag{5}$$

Substituting into the budget constraint:

$$\boxed{C_1^* = \frac{1}{1+\beta}\left(Y_1 + \frac{Y_2}{1+r}\right), \qquad C_2^* = \frac{\beta(1+r)}{1+\beta}\left(Y_1 + \frac{Y_2}{1+r}\right).} \tag{6}$$

Consumption is proportional to the *present value of lifetime resources* (permanent income).

Interest-rate changes for a borrower. A fall in r pivots the budget line around the endowment point. *Substitution effect*: C_1 becomes relatively cheaper $\Rightarrow C_1 \uparrow, C_2 \downarrow$. *Income effect*: a borrower is better off $\Rightarrow$ both $C_1 \uparrow, C_2 \uparrow$. Net: C_1 rises; for log utility the two effects on C_2 exactly cancel.

2.2 Life-cycle saving (Modigliani–Brumberg) and permanent income

Lifetime utility $U = \int_0^T u\big(C(t)\big)\,dt$ with $u' > 0$, $u'' < 0$; income $Y(t) = Y_0 + gt$ for $0 < t < R$ (working life) and $Y(t) = 0$ for $R \le t \le T$ (retirement); $r = 0$, no initial wealth, no uncertainty.

Lifetime budget constraint:

$$\int_0^T C(t)\,dt = \int_0^R (Y_0 + gt)\,dt = R\,Y_0 + \tfrac{1}{2}gR^2. \tag{7}$$

Consumption path. With $u'' < 0$, $r = 0$ and no impatience, utility maximization requires a *constant* consumption level (permanent-income logic). Spreading lifetime resources over T:

$$\boxed{\bar{C} = \frac{R}{T}\Big(Y_0 + \tfrac{1}{2}gR\Big).} \tag{8}$$

Wealth path. Saving $S(t) = Y(t) - \bar{C}$, wealth $W(t) = \int_0^t S(\tau)\,d\tau$:

$$W(t) = t\Big(Y_0 + \tfrac{1}{2}gt - \bar{C}\Big), \qquad 0 \le t < R, \tag{9}$$

$$W(R) = (T - R)\,\bar{C}, \qquad \text{at retirement}, \tag{10}$$

$$W(t) = (T - t)\,\bar{C}, \qquad R \le t \le T. \tag{11}$$

The consumer saves while income exceeds mean lifetime income and dissaves in retirement; wealth is maximized at retirement (*hump saving*) and reaches zero exactly at T.

2.3 Efficiency wages (Lecture 5)

Firm problem. N identical competitive firms maximize $\pi = Y - wL$ with production $Y = F(eL)$, $F' > 0$, $F'' < 0$, effort $e = e(w)$, $e' > 0$; $\bar{L}$ identical workers supply 1 unit of labor inelastically. First-order conditions:

$$F'\big(e(w)L\big) = \frac{w}{e(w)}, \tag{12}$$

$$\frac{w\,e'(w)}{e(w)} = 1. \tag{13}$$

Equation (13) is the **Solow condition**: at the optimum the elasticity of effort with respect to the wage equals 1. The wage satisfying it is the *efficiency wage*; firms never cut it even with involuntary unemployment.

Worked example (Lecture 5, eq. 12–17). Effort and market conditions:

$$e = \left(\frac{w - x}{x}\right)^{\beta} (w > x), \qquad x = (1 - bu)\,w_a, \qquad 0 < \beta < 1,\ b > 0, \tag{14}$$

where x measures labor-market conditions and w_a is the average wage. Applying (13):

$$\frac{w}{w - x} \cdot \beta = 1 \quad \Longrightarrow \quad \boxed{w^* = \frac{x}{1 - \beta} = \frac{1 - bu}{1 - \beta}\,w_a} \tag{15}$$

For small β, $w^* \approx (1+\beta)x$: the wage carries a premium of about β over x. In equilibrium $w = w_a$, so $1 = \frac{1-bu_{EQ}}{1-\beta}$, giving the **equilibrium unemployment rate**

$$\boxed{u_{EQ} = \frac{\beta}{b}} \tag{16}$$

which depends only on the effort-function parameters.

Wage rigidity. Let $u_{EQ} = 6\%$ and suppose unemployment rises to 9%. If the firm keeps the wage at w_a, effort rises (unemployment disciplines workers) and the unit labour cost is $C_{FIXED} = w_a/e(w_a, x')$; if it re-optimizes, $w' = x'/(1-\beta)$ and $C_{ADJ} = w'/e'$. The gain from cutting the wage is only a fraction of a percent (computed fully in Section 4.4) — so firms do not bother adjusting.

Shapiro–Stiglitz. Workers exert effort e or shirk (effort 0, detected at rate q, then fired); jobs break at rate b; unemployed find jobs at rate a; discount rate ρ. Value functions (derived in Section 4):

$$\rho V_E = w - e - b\,(V_E - V_U), \qquad \rho V_S = w - (b+q)\,(V_S - V_U), \qquad \rho V_U = a\,(V_E - V_U). \tag{17}$$

No shirking requires $V_E \geq V_S$, i.e. the employment rent $V_E - V_U \geq e/q$. The resulting **no-shirking condition (NSC)** with employment N:

$$\boxed{w = e + \frac{e}{q}\Big(\rho + b\,\frac{L}{L-N}\Big),} \tag{18}$$

upward-sloping in N; firms hire along labor demand $AF'(eL) = w/e$. Equilibrium is at the intersection: unemployment is *involuntary* and discipline comes from the threat of unemployment.

2.4 Real business cycles (Tutorial 5)

Aggregate production $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $0 < \alpha < 1$. Taking logs and subtracting the BGP:

$$\boxed{\hat{y}_t = \alpha \hat{k}_t + (1-\alpha)\big(\hat{A}_t + \hat{l}_t\big).} \tag{19}$$

Technology shock is AR(1): $\hat{A}_t = \rho_A \hat{A}_{t-1} + \varepsilon_{A,t}$, $\rho_A = 0.95$. After a one-time innovation $\varepsilon_{A,0}$:

$$\hat{A}_t = \varepsilon_{A,0}\,\rho_A^t, \qquad t \geq 0, \tag{20}$$

ρ_A is the *persistence*: the fraction of last period's deviation surviving into this period.

Calibrated policy rules (log-linearized, $\alpha = 1/3$):

$$\begin{aligned} \hat{c}_t &= 0.59\,\hat{k}_t + 0.38\,\hat{A}_t - 0.13\,\hat{g}_t, & \hat{l}_t &= -0.31\,\hat{k}_t + 0.35\,\hat{A}_t + 0.15\,\hat{g}_t, \\ \hat{k}_{t+1} &= 0.95\,\hat{k}_t + 0.08\,\hat{A}_t - 0.004\,\hat{g}_t. \end{aligned} \tag{21}$$

Propagation mechanisms: (1) persistence of the AR(1) technology process; (2) gradual endogenous capital accumulation — $\hat{k}_t$ is a *predetermined* state variable, so the shock's effect builds up over time.

Simplified RBC (Blanchard–Fischer). Infinite-lived representative agent maximizing $E\sum_{t=0}^{\infty}\frac{u(C_t)}{(1+\rho)^t}$ with $u(C_t) = C_t - \theta C_t^2$ ($u' > 0$); linear technology with additive disturbance $Y_t = AK_t + e_t$; no depreciation, $K_{t+1} = K_t + Y_t - C_t$; interest rate A; $A = \rho$; $e_t = \phi e_{t-1} + \varepsilon_t$, $|\phi| < 1$. The Euler equation (derived in Section 5.2):

$$1 - 2\theta E[C_{t+1}] = \frac{1+\rho}{1+A}\big(1 - 2\theta C_t\big) \xrightarrow{A=\rho} \boxed{E[C_{t+1}] = C_t} \tag{22}$$

consumption follows a **random walk**. The method of undetermined coefficients with $C_t = \alpha + \beta K_t + \gamma e_t$ yields

$$\boxed{\beta = A, \qquad \gamma = \frac{A}{1-\phi+A}, \qquad \alpha = 0,} \tag{23}$$

and a one-time shock $\varepsilon_t = 1 - \phi + A$ gives the impulse responses

$$\Delta C_{t+n} = A \;\; \forall n \geq 0, \qquad \Delta K_{t+n} = 1 - \phi^n, \qquad \Delta Y_{t+n} = A + \phi^n(1-\phi). \tag{24}$$

3 Tutorial 2 — Consumption: Complete Solutions

[Source: Tutorial 2 (Consumption), problems 1–3, with answer key]

3.1 Fisher's model with different saving and borrowing rates

Problem 1

The lecture analyzes Fisher's model when the consumer can save or borrow at one interest rate r, and the case in which he can save at r but cannot borrow at all. Consider the intermediate case: the consumer saves at rate r_s and borrows at rate r_b, where $r_s < r_b$.

(a) What is the budget constraint when he consumes less than his income in period one?

(b) What is the budget constraint when he consumes more than his income in period one?

(c) On a single graph show the two budget constraints; shade the combinations of (C_1, C_2) the consumer can choose.

(d) Add indifference curves and show three outcomes: saving, borrowing, neither.

(e) What determines first-period consumption in each of the three cases?

Solution. (a) A consumer who consumes less than his income in period one is a **saver**: his savings earn r_s and the second-period resource is $(Y_1 - C_1)(1 + r_s) + Y_2$. Dividing by $1 + r_s$:

$$C_1 + \frac{C_2}{1 + r_s} = Y_1 + \frac{Y_2}{1 + r_s}. \tag{25}$$

(b) A consumer who consumes more than his income in period one is a **borrower**: he borrows $C_1 - Y_1$ and repays $(C_1 - Y_1)(1 + r_b)$ next period:

$$C_1 + \frac{C_2}{1 + r_b} = Y_1 + \frac{Y_2}{1 + r_b}. \tag{26}$$

(c)–(d) Both lines pass through the endowment point $E = (Y_1, Y_2)$. The borrowing segment (right of E) is *steeper*: its slope is $-(1 + r_b) < -(1 + r_s)$; the saving segment (left of E) is flatter. The choice set is the area under the kinked frontier formed by the two segments (Figure 1).

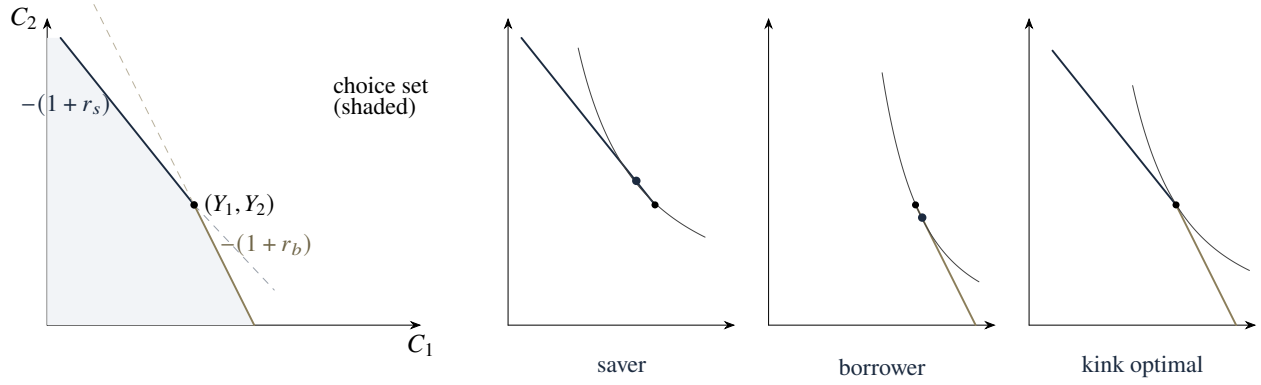


Figure 1: Fisher model with $r_s < r_b$: choice set (left) and the three tangency cases.

(e) **Saver**: the optimum is a tangency on the saver segment, so C_1 depends on present value at r_s, i.e. on $Y_1 + \frac{Y_2}{1+r_s}$ — both incomes and the rate r_s. **Borrower**: tangency on the borrowing segment, so C_1 depends on $Y_1 + \frac{Y_2}{1+r_b}$; borrowers discount future income more heavily because $r_b > r_s$. **Neither**: the highest attainable

indifference curve passes through the endowment point E, where the constraint has a kink, so $C_1 = Y_1$ — first-period consumption depends only on Y_1.

Answer

(a) $C_1 + \dfrac{C_2}{1+r_s} = Y_1 + \dfrac{Y_2}{1+r_s}$; (b) $C_1 + \dfrac{C_2}{1+r_b} = Y_1 + \dfrac{Y_2}{1+r_b}$. (e) Saver: C_1 depends on $Y_1 + \frac{Y_2}{1+r_s}$; borrower: on $Y_1 + \frac{Y_2}{1+r_b}$; at the kink: only on Y_1.

3.2 Log utility: Lagrangian, closed-form solution, and a numerical example

Problem 2

A representative consumer lives two periods with $U(C_1, C_2) = \ln C_1 + \beta \ln C_2$, $0 < \beta < 1$, and present-value budget constraint $C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}$, with r, Y_1, Y_2 exogenous.

(a) Set up the Lagrangian and derive the first-order conditions.

(b) Solve for the optimal C_1 and C_2.

(c) For $\beta = 0.9$, $Y_1 = 100$, $Y_2 = 200$, $r = 10\%$: calculate both consumption levels. Is he a saver or a borrower?

(d) If he earns nothing in period one but earns $(1 + 10\%) \cdot 100 + 200$ in period two, do the answers change?

(e) In the context of (d), what happens to the budget line when the interest rate falls? Which effect dominates — income or substitution?

Solution. (a) The Lagrangian with multiplier λ:

$$\mathcal{L} = \ln C_1 + \beta \ln C_2 + \lambda\Big(Y_1 + \frac{Y_2}{1+r} - C_1 - \frac{C_2}{1+r}\Big). \tag{27}$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial C_1} = 0: \quad \frac{1}{C_1} = \lambda, \tag{28}$$

$$\frac{\partial \mathcal{L}}{\partial C_2} = 0: \quad \frac{\beta}{C_2} = \frac{\lambda}{1+r}, \tag{29}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0: \quad C_1 + \frac{C_2}{1+r} = Y_1 + \frac{Y_2}{1+r}. \tag{30}$$

(b) Divide (29) by (28) to get the Euler equation $C_2 = (1+r)\beta C_1$. Substitute into (30):

$$C_1 + \frac{(1+r)\beta C_1}{1+r} = C_1(1+\beta) = Y_1 + \frac{Y_2}{1+r}, \tag{31}$$

hence

$$\boxed{C_1^* = \frac{1}{1+\beta}\Big(Y_1 + \frac{Y_2}{1+r}\Big), \qquad C_2^* = \frac{\beta(1+r)}{1+\beta}\Big(Y_1 + \frac{Y_2}{1+r}\Big).} \tag{32}$$

(c) Present value of lifetime resources:

$$Y_1 + \frac{Y_2}{1+r} = 100 + \frac{200}{1.1} = 281.82. \tag{33}$$

$$C_1^* = \frac{281.82}{1+0.9} = 148.33, \qquad C_2^* = (1.1)(0.9)(148.33) = 146.84. \tag{34}$$

Since $C_1 = 148.33 > Y_1 = 100$, he consumes more than current income: he is a **borrower** (he borrows 48.33 in period 1).

(d) New income path $Y_1' = 0$, $Y_2' = (1.1)(100) + 200 = 310$. Present value:

$$0 + \frac{310}{1.1} = 281.82 = Y_1 + \frac{Y_2}{1+r}. \tag{35}$$

The intertemporal budget line is *unchanged* — only its position relative to the endowment point changes (the endowment moves onto the C_2 axis). Answers to (a) and (b) are untouched, and (c) gives exactly the same consumption (148.33, 146.84): he simply **borrows more** (148.33 instead of 48.33) in period one.

(e) With the endowment at (0, 310), a fall in r **pivots the budget line around the endowment point** (the period-2 resource is unchanged). As a borrower he faces both effects: *substitution*: consuming today is cheaper, $C_1 \uparrow$, $C_2 \downarrow$; *income*: his real income rises (he repays less), $C_1 \uparrow$, $C_2 \uparrow$. Both raise C_1; they pull C_2 in opposite directions. With log utility they *exactly offset each other* for C_2. Re-running part (b) with $Y_1 = 0$, $Y_2 = 310$:

$$C_1 = \frac{310}{(1+\beta)(1+r)}, \qquad C_2 = \frac{\beta \cdot 310}{1+\beta}. \tag{36}$$

C_1 is decreasing in r: a drop in r **raises** C_1, while $C_2 = \frac{0.9 \cdot 310}{1.9} = 146.84$ does not depend on r at all. Numerically, $r : 10\% \to 5\%$ moves C_1 from 148.33 to 155.39 with C_2 fixed.

Answer

(b) $C_1^* = \frac{1}{1+\beta}\left(Y_1 + \frac{Y_2}{1+r}\right)$, $C_2^* = (1+r)\beta C_1^*$. (c) $C_1 = 148.33$, $C_2 = 146.84$ — a borrower. (d) No change: same PV = 281.82. (e) Budget line pivots around $(0, 310)$; substitution and income effects offset on C_2, so $C_1 \uparrow$, C_2 unchanged.

3.3 Life-cycle saving (Modigliani and Brumberg, 1954)

Problem 3

An individual lives from 0 to T with lifetime utility $U = \int_0^T u\big(C(t)\big)\,dt$, $u' > 0$, $u'' < 0$. Income is $Y_0 + gt$ for $0 < t < R$ and 0 for $R \le t \le T$, with $0 < R < T$. The interest rate is zero, initial wealth is zero, there is no uncertainty.

(a) What is the lifetime budget constraint?

(b) What is the utility-maximizing path of consumption $C(t)$?

(c) (Optional) What is the path of wealth $W(t)$?

Solution. (a) With zero initial wealth the present value of lifetime consumption cannot exceed the present value of lifetime income:

$$\int_0^T C(t)\,dt \;\le\; \int_0^T Y(t)\,dt. \tag{37}$$

Income is positive only during the working life, so

$$\int_0^T Y(t)\,dt = \int_0^R (Y_0 + gt)\,dt = \Big[Y_0 t + \tfrac{1}{2}gt^2\Big]_0^R = R\,Y_0 + \tfrac{1}{2}gR^2. \tag{38}$$

The **lifetime budget constraint** is

$$\boxed{\int_0^T C(t)\,dt \;\le\; R\,Y_0 + \tfrac{1}{2}gR^2.} \tag{39}$$

(b) Since $u'' < 0$ and both the interest rate and the discount rate are zero, the permanent-income logic of the PIH applies: utility maximization requires a **constant** consumption level $\bar{C}$. The budget constraint then spreads lifetime resources evenly over the lifetime T:

$$\bar{C} \cdot T = R\,Y_0 + \tfrac{1}{2} g R^2 \qquad \Longrightarrow \qquad \boxed{\bar{C} = \frac{R}{T}\Big(Y_0 + \tfrac{1}{2} g R\Big).} \tag{40}$$

(c) Saving is income minus consumption,

$$S(t) = Y(t) - \bar{C} = \begin{cases} Y_0 + gt - \bar{C}, & 0 < t < R, \\ -\bar{C}, & R \le t \le T, \end{cases} \tag{41}$$

and wealth is its integral, $W(t) = \int_0^t S(\tau)\, d\tau$. During the working life:

$$W(t) = \Big[Y_0 \tau + \tfrac{1}{2} g \tau^2 - \bar{C}\tau\Big]_0^t = t\Big(Y_0 + \tfrac{1}{2} g t - \bar{C}\Big), \qquad 0 \le t < R. \tag{42}$$

At retirement, using $\bar{C} = \frac{R}{T}\big(Y_0 + \frac{1}{2} g R\big)$, i.e. $R Y_0 + \frac{1}{2} g R^2 = \frac{T}{R}\bar{C} \cdot R = T\bar{C}$:

$$W(R) = R\Big(Y_0 + \tfrac{1}{2} g R - \bar{C}\Big) = T\bar{C} - R\bar{C} = (T - R)\,\bar{C}. \tag{43}$$

Intuition: retirement lasts $T - R$ years and must be financed entirely from accumulated wealth, so wealth at retirement equals $(T - R)\bar{C}$. During retirement saving is $-\bar{C}$, hence

$$W(t) = (T - R)\bar{C} - \int_R^t \bar{C}\, d\tau = (T - R)\bar{C} - (t - R)\bar{C} = (T - t)\,\bar{C}, \qquad R \le t \le T, \tag{44}$$

which declines linearly and reaches **zero exactly at** $t = T$: the consumer dies with no assets. This pattern (saving while income exceeds mean lifetime income, maximum wealth $(T-R)\bar{C}$ at retirement, then dissaving) is *hump saving* (Figure 2).

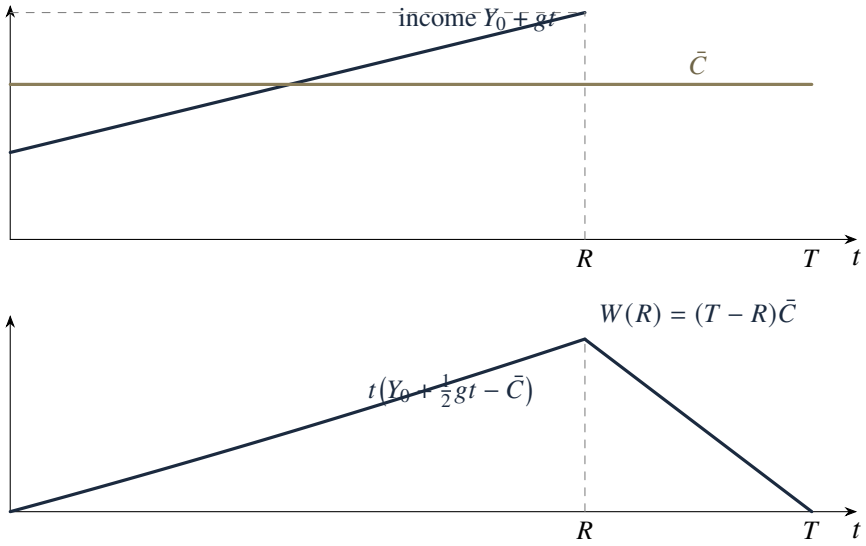


Figure 2: Life-cycle model: income rises to R then drops to zero while consumption is flat (top); wealth follows a hump, peaking at retirement and reaching zero at T (bottom).

Answer

(a) $\int_0^T C(t)\, dt \le R Y_0 + \frac{1}{2} g R^2$. (b) $\bar{C} = \frac{R}{T}\big(Y_0 + \frac{1}{2} g R\big)$ (constant). (c) $W(t) = t\big(Y_0 + \frac{1}{2} g t - \bar{C}\big)$ for $t < R$; $W(R) = (T - R)\bar{C}$; $W(t) = (T - t)\bar{C}$ for $R \le t \le T$.

4 Tutorial 4 — Unemployment: Complete Solutions

[Source: Tutorial 4 (Unemployment), problems 1–3, with answer key]

4.1 Shapiro–Stiglitz value functions

Problem 1

Following the lecture notes (pages 24–26), derive the value functions for a worker in state S (employed and shirking) and state U (unemployed), i.e. equations (28) and (29) of the notes.

Solution. **Shirker (S).** A shirking worker receives the wage w (no effort cost) but is caught shirking at hazard rate q and fired; independently, the job breaks at rate b. Over a small interval Δt he is still employed Δt later with probability $e^{-(b+q)\Delta t}$, otherwise unemployed. Discounting at ρ:

$$V_S(\Delta t) = \int_0^{\Delta t} e^{-(b+q)t}\, e^{-\rho t} w \, dt + e^{-\rho\Delta t}\Big[e^{-(b+q)\Delta t}\, V_S(\Delta t) + \big(1 - e^{-(b+q)\Delta t}\big)\, V_U(\Delta t)\Big] \tag{45}$$

$$= \frac{1}{\rho + b + q}\Big(1 - e^{-(\rho+b+q)\Delta t}\Big) w + e^{-\rho\Delta t}\Big[e^{-(b+q)\Delta t}\, V_S(\Delta t) + \big(1 - e^{-(b+q)\Delta t}\big)\, V_U(\Delta t)\Big]. \tag{46}$$

Collecting the V_S terms, $V_S(\Delta t)\big(1 - e^{-(\rho+b+q)\Delta t}\big) = \big(1 - e^{-(\rho+b+q)\Delta t}\big)\frac{w}{\rho+b+q} + e^{-\rho\Delta t}\big(1 - e^{-(b+q)\Delta t}\big)V_U$, so

$$V_S(\Delta t) = \frac{1}{\rho + b + q}\, w + \frac{e^{-\rho\Delta t}\big(1 - e^{-(b+q)\Delta t}\big)}{1 - e^{-(\rho+b+q)\Delta t}}\, V_U. \tag{47}$$

Let $\Delta t \to 0$ and apply L'Hôpital's rule to the ratio, $\dfrac{e^{-\rho\Delta t}(1 - e^{-(b+q)\Delta t})}{1 - e^{-(\rho+b+q)\Delta t}} \to \dfrac{(b+q)\Delta t}{(\rho+b+q)\Delta t} = \dfrac{b+q}{\rho+b+q}$:

$$V_S = \frac{1}{\rho + b + q}\Big[w + (b+q)\, V_U\Big] \qquad \Longleftrightarrow \qquad \boxed{\rho V_S = w - (b+q)\,(V_S - V_U).} \tag{48}$$

Unemployed (U). An unemployed worker has zero income and finds a job within Δt with probability $a\Delta t$:

$$V_U(\Delta t) = \int_0^{\Delta t} 0 \, dt + e^{-\rho\Delta t}\Big[e^{-a\Delta t}\, V_U(\Delta t) + \big(1 - e^{-a\Delta t}\big)\, V_E(\Delta t)\Big] \tag{49}$$

$$= e^{-(a+\rho)\Delta t}\, V_U(\Delta t) + e^{-\rho\Delta t}\big(1 - e^{-a\Delta t}\big)\, V_E(\Delta t). \tag{50}$$

Collecting terms and taking $\Delta t \to 0$:

$$V_U(\Delta t)\big(1 - e^{-(a+\rho)\Delta t}\big) = e^{-\rho\Delta t}\big(1 - e^{-a\Delta t}\big)V_E \implies \frac{V_U}{V_E} = \lim_{\Delta t\to 0}\frac{a\Delta t}{(a+\rho)\Delta t} = \frac{a}{a+\rho}, \tag{51}$$

$$\boxed{V_U = \frac{a}{a+\rho}\, V_E \qquad \Longleftrightarrow \qquad \rho V_U = a\,(V_E - V_U).} \tag{52}$$

(The printed key shows $V_U = \frac{\rho}{a+\rho}V_E$ — a typo; the flow equation $\rho V_U = a(V_E - V_U)$ given there is correct and implies $V_U = \frac{a}{a+\rho}V_E$.)

No shirking. The non-shirker's value function is $\rho V_E = w - e - b(V_E - V_U)$. Firms prevent shirking iff $V_E \geq V_S$; by (48) this requires the employment rent $V_E - V_U \geq e/q$, which is the source of the NSC used in Problem 2.

Answer

$\rho V_S = w - (b+q)(V_S - V_U)$ and $\rho V_U = a(V_E - V_U)$, i.e. $V_S = \frac{w+(b+q)V_U}{\rho+b+q}$, $V_U = \frac{a}{a+\rho}V_E$.

4.2 Comparative statics in the Shapiro–Stiglitz model

Problem 2

Describe how each of the following affects equilibrium employment and the wage in the Shapiro–Stiglitz model: (a) an increase in workers' discount rate ρ; (b) an increase in the job breakup rate b; (c) a positive multiplicative shock to production ($A \uparrow$ in $AF(L)$); (d) an increase in the size of the labor force L.

Solution. Equilibrium is the intersection of two curves in (N, w) space — employment N, wage w:

$$\textbf{NSC:} \quad w = e + \frac{e}{q}\left(\rho + b\,\frac{L}{L-N}\right) \qquad \text{(upward-sloping, vertical asymptote at } N = L\text{)}, \tag{53}$$

$$\textbf{Labor demand:} \quad AF'(eL) = \frac{w}{e} \qquad \text{(downward-sloping)} : \tag{54}$$

firms hire effective labor eL until its marginal product equals its marginal cost w/e.

(a) ρ **up.** From (53), at any N the wage needed to deter shirking rises, so the NSC shifts *up*. Intuition: workers discount the future more, so being caught and fired matters less; the employment rent shrinks and only a higher wage recreates it. Labor demand (54) is unaffected. **Employment falls, wage rises** (Figure 3, left).

(b) b **up.** Same mechanism: with jobs breaking more often anyway, the value of holding a job is lower, shirking is less costly, and the NSC shifts *up* — at a given N firms must pay more to deter shirking. Labor demand unchanged. **Employment falls, wage rises** (same figure as (a)).

(c) A **up.** The technology shock shifts *labor demand* right; the NSC is untouched. **Employment rises, wage rises.** Note the contrast with a competitive spot market: with inelastic labor supply and no efficiency wages, $A \uparrow$ would raise only the wage, not employment — efficiency wages make employment respond to technology.

(d) L **up.** The vertical labor-supply line moves right and the NSC shifts *down*: at a given N, $L - N$ is larger, so a shirker who is fired stays unemployed longer, the expected cost of shirking rises, and firms can deter shirking with a lower wage. Labor demand is unaffected. **Employment rises, wage falls** (Figure 3, right).

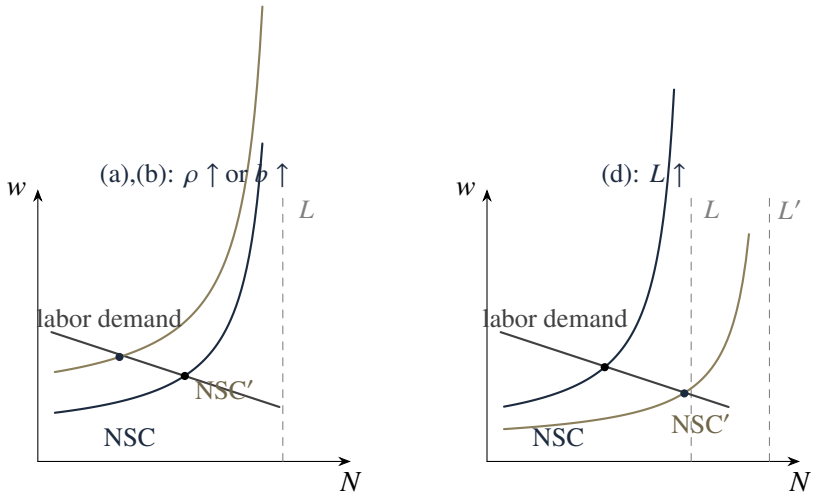


Figure 3: Shapiro–Stiglitz comparative statics. Left: $\rho \uparrow$ or $b \uparrow$ shifts the NSC up — employment falls, wage rises. Right: $L \uparrow$ shifts the NSC down — employment rises, wage falls.

Answer

(a) NSC up: employment ↓, wage ↑. (b) NSC up: employment ↓, wage ↑. (c) Labor demand right: employment ↑, wage ↑. (d) NSC down: employment ↑, wage ↓.

4.3 Union wage premiums and efficiency wages (Summers, 1988)

Problem 3

In the efficiency-wage model of Lecture 5 (eqs. 12–17), a fraction f of workers belong to unions that obtain a wage exceeding the nonunion wage by proportion μ: $w_u = (1+\mu)w_n$. The average wage is $w_a = f\,w_u + (1-f)w_n$, and nonunion employers set wages freely, so $w_n = \frac{1-bu}{1-\beta}\,w_a$.

(a) Find the equilibrium unemployment rate in terms of β, b, f, μ.

(b) For $\mu = f = 0.15$, $\beta = 0.06$, $b = 1$: compute u. By what proportion is the cost of effective labor higher in the union sector?

(c) Repeat (b) for $\beta = 0.03$, $b = 0.5$.

Solution. (a) Substitute $w_a = f w_u + (1-f)w_n$ and $w_u = (1+\mu)w_n$ into the nonunion wage rule:

$$w_n = \frac{1-bu}{1-\beta}\Big[f(1+\mu)w_n + (1-f)w_n\Big] = \frac{1-bu}{1-\beta}\,(1+\mu f)\,w_n. \tag{55}$$

Cancel w_n (positive in equilibrium):

$$(1-bu)(1+\mu f) = 1-\beta \implies 1-bu = \frac{1-\beta}{1+\mu f} \implies \boxed{u = \frac{\beta+\mu f}{b\,(1+\mu f)}.} \tag{56}$$

With $f = 0$ this collapses to the standard result $u_{EQ} = \beta/b$.

(b) Numerically:

$$u = \frac{0.06 + (0.15)(0.15)}{1 + (0.15)(0.15)} = \frac{0.06 + 0.0225}{1.0225} = \frac{0.0825}{1.0225} = 0.0807 \approx \mathbf{8.1}\%, \tag{57}$$

higher than the 6% of the standard model ($\beta/b = 0.06$) without a union sector.

Effective-labour cost. Effort is $e = \left(\frac{w-x}{x}\right)^{\beta}$ with $x = (1-bu)w_a$. For union workers, $w_u = (1+\mu)\frac{1-bu}{1-\beta}w_a$, so

$$\frac{w_u - x}{x} = \frac{(1+\mu)}{1-\beta} - 1 = \frac{\mu+\beta}{1-\beta} \qquad \Longrightarrow \qquad e_u = \left(\frac{\mu+\beta}{1-\beta}\right)^{\beta}; \tag{58}$$

for nonunion workers, $w_n = \frac{1-bu}{1-\beta}w_a$ gives

$$e_n = \left(\frac{\beta}{1-\beta}\right)^{\beta}. \tag{59}$$

Unit costs of effective labor are w_u/e_u and w_n/e_n; their ratio:

$$\frac{w_u/e_u}{w_n/e_n} = (1+\mu)\left(\frac{\beta}{\mu+\beta}\right)^{\beta}. \tag{60}$$

With $\mu = 0.15$, $\beta = 0.06$:

$$\frac{w_u/e_u}{w_n/e_n} = 1.15\left(\frac{0.06}{0.21}\right)^{0.06} = 1.15\,(0.2857)^{0.06} = 1.15 \cdot 0.9276 = \mathbf{1.0667}. \tag{61}$$

Although the wage in the union sector is 15% higher, the cost of *effective* labor is only about 6.7% higher: union workers exert more effort because they are paid above the market.

(c) For $\beta = 0.03$ (the cost ratio is independent of b):

$$\frac{w_u/e_u}{w_n/e_n} = 1.15\left(\frac{0.03}{0.18}\right)^{0.03} = 1.15\,(0.1667)^{0.03} = 1.15 \cdot 0.9476 = \mathbf{1.0898}. \tag{62}$$

(The printed key writes the exponent as 0.06 and "substituting $\beta = 0.06$" — typos; the value 1.0898 corresponds to the exponent $\beta = 0.03$.) With the elasticity of effort lower (β smaller), effort responds less to the wage premium, so the union-sector cost of effective labor is now 9.0% higher — more than in (b).

Answer

(a) $u = \dfrac{\beta + \mu f}{b(1+\mu f)}$. (b) $u \approx 8.1\%$; cost ratio $= 1.15\left(\frac{0.06}{0.21}\right)^{0.06} = 1.0667$. (c) $1.15\left(\frac{0.03}{0.18}\right)^{0.03} = 1.0898$.

4.4 Wage rigidity in the efficiency-wage model (Lecture 5, eqs. 18–19)

Problem 4

Let $u_{EQ} = 6\%$ and suppose unemployment rises to 9%. Compute the change in the unit cost of effective labor if the firm (i) keeps the wage unchanged, (ii) optimally adjusts the wage, for the calibrations (1) $\beta = 0.06$, $b = 1$ and (2) $\beta = 0.03$, $b = 0.5$. How large is the incentive to cut wages?

Solution. Normalize $w_a = 1$. At the equilibrium $u_{EQ} = \beta/b$, effort and cost are

$$e_0 = \left(\frac{\beta}{1-\beta}\right)^{\beta}, \qquad C_0 = \frac{w_a}{e_0}. \tag{63}$$

After unemployment rises to u_1, market conditions worsen to $x' = (1 - bu_1)w_a$. **(i) Fixed wage:** the firm keeps $w = w_a$; workers' effort *rises* (unemployment disciplines), and the unit cost becomes

$$C_{FIXED} = \frac{w_a}{e(w_a, x')}, \qquad e(w_a, x') = \left(\frac{w_a - x'}{x'}\right)^{\beta}. \tag{64}$$

(ii) Adjusted wage: the firm re-optimizes to $w' = x'/(1-\beta)$, which restores the original effort-cost ratio $(w' - x')/x' = \beta/(1-\beta)$, so $e' = e_0$ and

$$C_{ADJ} = \frac{w'}{e'} = \frac{x'/(1-\beta)}{e_0}. \tag{65}$$

The numbers (changes measured relative to C_0; the gain from cutting is the difference $\Delta C_{FIXED} - \Delta C_{ADJ}$):

Calibration	ΔC_{FIXED}	ΔC_{ADJ}	Gain from cutting
(1) $\beta = 0.06$, $b = 1$	−2.6%	−3.2%	0.6 pp
(2) $\beta = 0.03$, $b = 0.5$	−1.3%	−1.5%	0.2 pp

A 3-percentage-point rise in unemployment lowers unit costs by only 1.3–3.2%, and *optimally* adjusting the wage saves at most a further 0.2–0.6 percentage points. The incentive to renegotiate wages is tiny — this is why wages are **rigid** downward in the efficiency-wage model even in the face of large unemployment.

Answer

$\Delta C_{FIXED} = -2.6\%$, $\Delta C_{ADJ} = -3.2\%$, gain 0.6 pp (case 1); -1.3%, -1.5%, 0.2 pp (case 2) — wage cutting is not worth the adjustment.

5 Tutorial 5 — Real-Business-Cycle Theory: Complete Solutions

[Source: Tutorial 5 (Real-Business-Cycle Theory), problems 1–2, with answer key]

5.1 Technology shocks and macroeconomic fluctuations in the RBC model

Problem 1

Consider the RBC model of the lecture notes with aggregate production $Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}$, $0 < \alpha < 1$, and $\hat{x}_t \equiv \ln x_t - \ln \bar{x}_t$ denoting deviations from the balanced growth path.

(a) Technology follows $\hat{A}_t = \rho_A \hat{A}_{t-1} + \varepsilon_{A,t}$, $\rho_A = 0.95$. The economy starts on its BGP; at $t = 0$ an unexpected positive innovation of 1 percent arrives ($\varepsilon_{A,0} = 0.01$) and there are no further innovations. (i) Compute $\hat{A}_0, \hat{A}_1, \hat{A}_2$. (ii) Write $\hat{A}_t$ for any $t \geq 0$. (iii) Explain what ρ_A means.

(b) The calibrated model gives the log-linearized rules $\hat{c}_t = 0.59\hat{k}_t + 0.38\hat{A}_t - 0.13\hat{g}_t$, $\hat{l}_t = -0.31\hat{k}_t + 0.35\hat{A}_t + 0.15\hat{g}_t$, $\hat{k}_{t+1} = 0.95\hat{k}_t + 0.08\hat{A}_t - 0.004\hat{g}_t$, with $\alpha = 1/3$, $\hat{k}_0 = 0$, $\hat{g}_t = 0$ always. (i) Starting from the Cobb–Douglas production function, derive $\hat{y}_t$. (ii) Compute the responses of C, L, Y at $t = 0$ and of K at $t = 1$. (iii) Continue for $t = 1, 2$ and complete the table of percentage deviations.

(c) Using (b), explain: (i) why capital responds with a lag; (ii) why consumption can keep rising as the shock fades; (iii) the two mechanisms generating persistent fluctuations.

Solution. (a) With $\hat{A}_{-1} = 0$ and the single innovation $\varepsilon_{A,0} = 0.01$:

$$\hat{A}_0 = \rho_A \underbrace{\hat{A}_{-1}}_{0} + \varepsilon_{A,0} = 0.01, \tag{66}$$

$$\hat{A}_1 = \rho_A \hat{A}_0 = 0.95(0.01) = 0.0095, \tag{67}$$

$$\hat{A}_2 = \rho_A \hat{A}_1 = 0.95(0.0095) = 0.009025. \tag{68}$$

(ii) Absent further innovations the recursion unfolds as a geometric sequence:

$$\boxed{\hat{A}_t = 0.01\,(0.95)^t, \qquad t \geq 0.} \tag{69}$$

The technology deviations in periods $0, 1, 2$ are 1%, 0.95%, 0.9025%. (iii) The persistence parameter $\rho_A = 0.95$ means that **95 percent of the previous period's deviation remains in the current period**: the shock fades gradually (×0.95 each period) rather than disappearing immediately.

(b)(i) Take logs of the Cobb–Douglas production function, $\ln Y_t = \alpha \ln K_t + (1-\alpha)(\ln A_t + \ln L_t)$, and subtract the balanced-growth-path expression:

$$\boxed{\hat{y}_t = \alpha \hat{k}_t + (1-\alpha)\big(\hat{A}_t + \hat{l}_t\big) \xrightarrow{\alpha = 1/3} \hat{y}_t = \tfrac{1}{3}\hat{k}_t + \tfrac{2}{3}\big(\hat{A}_t + \hat{l}_t\big).} \tag{70}$$

(b)(ii) At $t = 0$: $\hat{k}_0 = 0$, $\hat{A}_0 = 0.01$:

$$\hat{c}_0 = 0.59(0) + 0.38(0.01) = 0.0038 = 0.38\%, \qquad \hat{l}_0 = -0.31(0) + 0.35(0.01) = 0.0035 = 0.35\%, \tag{71}$$

$$\hat{y}_0 = \tfrac{1}{3}(0) + \tfrac{2}{3}(0.01 + 0.0035) = 0.009 = 0.90\%, \quad \hat{k}_1 = 0.95(0) + 0.08(0.01) = 0.0008 = 0.08\%. \tag{72}$$

(b)(iii) At $t = 1$: $\hat{A}_1 = 0.0095$, $\hat{k}_1 = 0.0008$:

$$\hat{c}_1 = 0.59(0.0008) + 0.38(0.0095) = 0.000472 + 0.00361 = 0.004082 = 0.4082\%, \tag{73}$$

$$\hat{l}_1 = -0.31(0.0008) + 0.35(0.0095) = -0.000248 + 0.003325 = 0.003077 = 0.3077\%, \tag{74}$$

$$\hat{y}_1 = \tfrac{1}{3}(0.0008) + \tfrac{2}{3}(0.0095 + 0.003077) = 0.0086513 = 0.8651\%, \tag{75}$$

$$\hat{k}_2 = 0.95(0.0008) + 0.08(0.0095) = 0.00076 + 0.00076 = 0.00152 = 0.152\%. \tag{76}$$

At $t = 2$: $\hat{A}_2 = 0.009025$, $\hat{k}_2 = 0.00152$:

$$\hat{c}_2 = 0.59(0.00152) + 0.38(0.009025) = 0.0008968 + 0.0034295 = 0.0043263 = 0.4326\%, \tag{77}$$

$$\hat{l}_2 = -0.31(0.00152) + 0.35(0.009025) = -0.0004712 + 0.0031588 = 0.0026876 = 0.2688\%, \tag{78}$$

$$\hat{y}_2 = \tfrac{1}{3}(0.00152) + \tfrac{2}{3}(0.009025 + 0.0026876) = 0.0083150 = 0.8315\%. \tag{79}$$

Variable (% deviations from BGP)	$t=0$	$t=1$	$t=2$
Technology, $\hat{A}_t$	1.0000	0.9500	0.9025
Capital, $\hat{k}_t$	0.0000	0.0800	0.1520
Consumption, $\hat{c}_t$	0.3800	0.4082	0.4326
Labor, $\hat{l}_t$	0.3500	0.3077	0.2688
Output, $\hat{y}_t$	0.9000	0.8651	0.8315

All entries are *levels* of percentage deviation from the BGP, not growth rates between consecutive periods.

(c)(i) **Capital is a predetermined state variable.** K_{t+1} is chosen at date t through investment and installed for date $t+1$; it cannot jump when the shock arrives, so its response appears only in subsequent periods ($\hat{k}_0 = 0$, $\hat{k}_1 = 0.08\%$, $\hat{k}_2 = 0.152\%$).

(c)(ii) The policy rule $\hat{c}_t = 0.59\hat{k}_t + 0.38\hat{A}_t$ is *positive* in both state variables. Although $\hat{A}_t$ declines ($1\% \to 0.95\% \to 0.9025\%$), the capital stock accumulates ($0 \to 0.08\% \to 0.152\%$), and its positive contribution more than offsets the fading technology contribution over the first periods: $0.59 \times 0.08\% = 0.047\%$ added at $t=1$ against a technology loss of only 0.019%.

(c)(iii) The two propagation mechanisms are: **(1) persistence in the AR(1) technology process** — $\rho_A < 1$ makes the technology deviation fade slowly; **(2) gradual endogenous capital accumulation** — investment carries the effects of the shock into future periods. Together they turn a one-time 1% innovation into a prolonged fluctuation of output, consumption, and labor.

Answer

(a) $\hat{A}_t = 0.01(0.95)^t$: 1.000%, 0.950%, 0.9025%. (b) Table above; $\hat{y}_t = \frac{1}{3}\hat{k}_t + \frac{2}{3}(\hat{A}_t + \hat{l}_t)$. (c) Capital predetermined; consumption rule positive in $\hat{k}, \hat{A}$; propagation = AR(1) persistence + capital accumulation.

5.2 A simplified RBC model with additive technology shocks (Blanchard–Fischer)

Problem 2

The population is constant and infinitely lived. The representative individual maximizes the expected discounted stream $E\sum_{t=0}^{\infty} u(C_t)/(1+\rho)^t$ with $\rho > 0$ and $u(C_t) = C_t - \theta C_t^2$, where $u' > 0$ over the relevant range. Output is linear in capital plus an additive disturbance: $Y_t = AK_t + e_t$. There is no depreciation, so $K_{t+1} = K_t + Y_t - C_t$, and the interest rate is A. Assume $A = \rho$. The disturbance is AR(1): $e_t = \phi e_{t-1} + \varepsilon_t$, $|\phi| < 1$, ε_t i.i.d. mean zero.

(a) Find the first-order condition (Euler equation) relating C_t and expectations of C_{t+1}.

(b) Guess $C_t = \alpha + \beta K_t + \gamma e_t$. What is K_{t+1} as a function of K_t and e_t?

(c) What values of α, β, γ satisfy the first-order condition for all K_t and e_t?

(d) If e (and ε) equal 0 until period t, when a one-time positive realization $\varepsilon_t = 1 - \phi + A$ occurs (and $\varepsilon = 0$ again afterwards), what are the effects on the paths of Y, K, C?

Solution. (a) The accumulation and production functions imply

$$C_t = (1+A)K_t - K_{t+1} + e_t. \tag{80}$$

Let $V(K_t)$ be the optimal value function. The Bellman equation is

$$V(K_t) = \max_{K_{t+1}}\Big[(1+A)K_t - K_{t+1} + e_t - \theta\big((1+A)K_t - K_{t+1} + e_t\big)^2 + \frac{1}{1+\rho}\,E_t V(K_{t+1})\Big]. \tag{81}$$

FOC (derivative w.r.t. K_{t+1}, using $u'(C_t) = 1 - 2\theta C_t$):

$$1 - 2\theta\big[(1+A)K_t + e_t - K_{t+1}\big] + \frac{1}{1+\rho}\,E_t V'(K_{t+1}) = 0, \tag{82}$$

i.e. after rearrangement,

$$E_t V'(K_{t+1}) = (1+\rho)\,\big(1 - 2\theta C_t\big). \tag{83}$$

Envelope theorem: $V'(K_t) = (1+A)\big(1-2\theta[(1+A)K_t + e_t - K_{t+1}]\big) = (1+A)(1-2\theta C_t)$, shifted one period ahead:

$$V'(K_{t+1}) = (1+A)\big(1 - 2\theta C_{t+1}\big). \tag{84}$$

Combining (83) and (84) gives the intertemporal Euler equation:

$$1 - 2\theta\,E[C_{t+1}] = \frac{1+\rho}{1+A}\big(1 - 2\theta C_t\big). \tag{85}$$

Using $A = \rho$ the fraction equals 1, so $1 - 2\theta E[C_{t+1}] = 1 - 2\theta C_t$:

$$\boxed{E[C_{t+1}] = C_t} \tag{86}$$

consumption follows a random walk: the expected value of tomorrow's consumption equals today's realized consumption, regardless of anything known at t.

(b) Substitute the guess $C_t = \alpha + \beta K_t + \gamma e_t$ and $Y_t = AK_t + e_t$ into $K_{t+1} = K_t + Y_t - C_t$:

$$K_{t+1} = K_t + AK_t + e_t - \alpha - \beta K_t - \gamma e_t, \tag{87}$$

$$\boxed{K_{t+1} = -\alpha + (1 + A - \beta)\,K_t + (1-\gamma)\,e_t.} \tag{88}$$

(c) Write the Euler condition $E[C_{t+1}] = C_t$ for the guessed form. Since K_t, e_t are known at t and $E_t[e_{t+1}] = \phi e_t$:

$$E_t\big[\alpha + \beta K_{t+1} + \gamma e_{t+1}\big] = \alpha + \beta K_t + \gamma e_t, \tag{89}$$

$$\alpha + \beta E_t[K_{t+1}] + \gamma\phi e_t = \alpha + \beta K_t + \gamma e_t, \tag{90}$$

$$\alpha + \beta\big[-\alpha + (1 + A - \beta)K_t + (1-\gamma)e_t\big] + \gamma\phi e_t = \alpha + \beta K_t + \gamma e_t. \tag{91}$$

Collecting terms:

$$\alpha(1-\beta) + \beta(1 + A - \beta)\,K_t + \big[\beta + \gamma(\phi - \beta)\big]\,e_t = \alpha + \beta K_t + \gamma e_t. \tag{92}$$

For this to hold for *all* K_t and e_t, match coefficients:

$$\begin{cases} \beta(1 + A - \beta) = \beta, \\ \beta + \gamma(\phi - \beta) = \gamma, \\ \alpha(1-\beta) = \alpha, \end{cases} \tag{93}$$

Solving:

$$\beta(1 + A - \beta) = \beta \implies \beta = A \quad (\beta \neq 0), \tag{94}$$

$$\beta + \gamma\phi - \gamma\beta = \gamma \implies \beta = \gamma(1 - \phi + \beta) \implies \gamma = \frac{A}{1 - \phi + A}, \tag{95}$$

$$\alpha(1-\beta) = \alpha \implies \alpha(-\beta) = 0 \implies \alpha = 0 \quad (\beta = A \neq 0). \tag{96}$$

$$\boxed{\beta = A, \qquad \gamma = \frac{A}{1 - \phi + A}, \qquad \alpha = 0} \tag{97}$$

(The degenerate solution $\beta = \gamma = 0$ with unrestricted α is ignored — it makes less economic sense.) Hence

$$C_t = AK_t + \frac{A}{1-\phi+A}\, e_t, \qquad K_{t+1} = K_t + \frac{1-\phi}{1-\phi+A}\, e_t. \tag{98}$$

(d) From (98), K_t depends on last period's capital and last period's e, both unaffected by a shock arriving at t: $\Delta K_t = 0$. With $\Delta e_t = \varepsilon_t = 1 - \phi + A$:

$$\Delta Y_t = A\,\Delta K_t + \Delta e_t = 1 - \phi + A, \tag{99}$$

$$\Delta C_t = A\,\Delta K_t + \frac{A}{1-\phi+A}\,\Delta e_t = \frac{A}{1-\phi+A}\,(1-\phi+A) = A. \tag{100}$$

Output jumps by $1 - \phi + A$; consumption jumps by A. The remainder of the output increase, $(1 - \phi)$, is invested: since output rose by $(1 - \phi + A)$ but consumption only by A, capital next period rises by

$$\Delta K_{t+1} = \frac{1-\phi}{1-\phi+A}\,\Delta e_t = 1 - \phi. \tag{101}$$

At $t + 1$ the AR(1) form keeps e above its old path even though $\varepsilon_{t+1} = 0$:

$$\Delta e_{t+1} = \phi\,\Delta e_t = \phi\,(1-\phi+A), \tag{102}$$

$$\Delta Y_{t+1} = A\,\Delta K_{t+1} + \Delta e_{t+1} = A(1-\phi) + \phi(1-\phi+A) = A + \phi(1-\phi), \tag{103}$$

$$\Delta C_{t+1} = A\,\Delta K_{t+1} + \frac{A}{1-\phi+A}\,\Delta e_{t+1} = A(1-\phi) + \frac{A}{1-\phi+A}\,\phi(1-\phi+A) = A(1-\phi) + A\phi = A. \tag{104}$$

Consumption has **no further dynamics**: it stays permanently higher by A. Repeating for $t + 2$:

$$\Delta e_{t+2} = \phi^2(1-\phi+A), \qquad \Delta K_{t+2} = \Delta K_{t+1} + \frac{1-\phi}{1-\phi+A}\,\Delta e_{t+1} = 1 - \phi^2, \tag{105}$$

$$\Delta Y_{t+2} = A\,(1-\phi^2) + \phi^2(1-\phi+A) = A + \phi^2(1-\phi), \qquad \Delta C_{t+2} = A. \tag{106}$$

The pattern for a one-time shock $\varepsilon_t = 1 - \phi + A$:

$$\boxed{\Delta C_{t+n} = A \;\; \forall n \geq 0; \qquad \Delta K_{t+n} = 1 - \phi^n; \qquad \Delta Y_{t+n} = A + \phi^n(1-\phi).} \tag{107}$$

Capital rises to a permanently higher level (by 1); output overshoots to $1 - \phi + A$ and then converges down to a level A higher than before; consumption jumps by A immediately and stays there. The dynamics of Y and K depend on ϕ (Figure 4):

- $\phi = 0$ (no persistence): no dynamics after $t + 1$ — from then on capital is higher by 1 and output by A, forever.
- $0 < \phi < 1$: capital rises by $(1 - \phi)$ the period after the shock and then increases further each period, asymptotically approaching 1; output spikes to $1 - \phi + A$ and decays monotonically toward A.
- $-1 < \phi < 0$: capital and output *oscillate* around their new long-run levels, alternating above and below, and gradually settle at 1 and A higher respectively.

Answer

(a) $E[C_{t+1}] = C_t$ — random walk. (b) $K_{t+1} = -\alpha + (1 + A - \beta)K_t + (1 - \gamma)e_t$. (c) $\beta = A$, $\gamma = \frac{A}{1-\phi+A}$, $\alpha = 0$. (d) $\Delta C_{t+n} = A$ (no dynamics), $\Delta K_{t+n} = 1 - \phi^n$, $\Delta Y_{t+n} = A + \phi^n(1 - \phi)$.

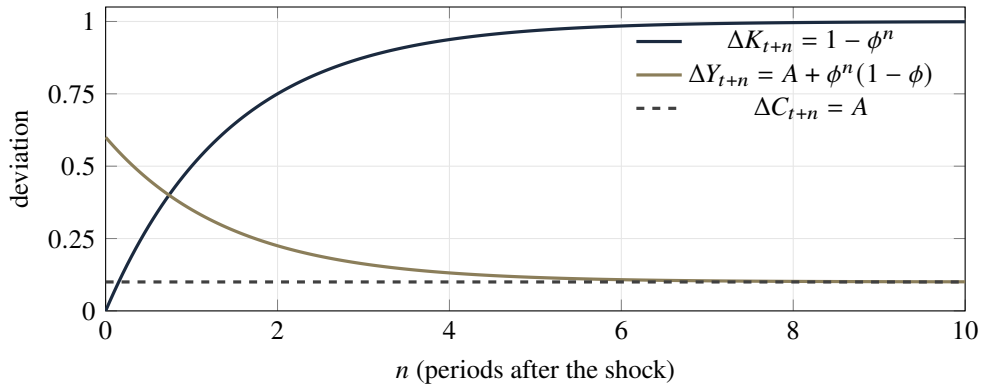


Figure 4: Blanchard–Fischer impulse responses to the one-time shock $\varepsilon_t = 1 - \phi + A$ with $\phi = 0.5$, $A = 0.1$: output overshoots and decays to A, capital builds up gradually to 1, consumption jumps by A and stays there.

6 Exam Crib Sheet — All Key Results on One Page

Two-period consumption (Fisher). Budget (saver / borrower): $C_1 + \frac{C_2}{1+r_{s,b}} = Y_1 + \frac{Y_2}{1+r_{s,b}}$; both lines pass through (Y_1, Y_2); feasible set lies between them. Euler: $u'(C_1) = \beta(1+r)\,u'(C_2)$. Log utility: $C_2 = (1+r)\beta C_1$; $C_1^* = \frac{1}{1+\beta}\big(Y_1 + \frac{Y_2}{1+r}\big)$, $C_2^* = \frac{\beta(1+r)}{1+\beta}\big(Y_1 + \frac{Y_2}{1+r}\big)$. Borrower, $r \downarrow$: substitution ($C_1 \uparrow$) and income ($C_1 \uparrow, C_2 \uparrow$) effects — C_1 rises; with log utility C_2 is unchanged.

Life-cycle saving. $\int_0^T C\,dt = RY_0 + \frac{1}{2}gR^2$; $\bar{C} = \frac{R}{T}\big(Y_0 + \frac{1}{2}gR\big)$ (constant); $S(t) = Y(t) - \bar{C}$; $W(t) = t\big(Y_0 + \frac{1}{2}gt - \bar{C}\big)$ for $t < R$; $W(R) = (T-R)\bar{C}$; $W(t) = (T-t)\bar{C}$.

Efficiency wages. Firm: $\pi = F(eL) - wL$; FOC: $F'(eL) = w/e$; Solow: $we'(w)/e(w) = 1$. Example: $e = \big(\frac{w-x}{x}\big)^\beta$, $x = (1-bu)w_a \Rightarrow w^* = \frac{x}{1-\beta} = \frac{1-bu}{1-\beta}w_a$, $u_{EQ} = \beta/b$. Efforts: $e_u = \big(\frac{\mu+\beta}{1-\beta}\big)^\beta$, $e_n = \big(\frac{\beta}{1-\beta}\big)^\beta$; union premium: $u = \frac{\beta+\mu f}{b(1+\mu f)}$; cost ratio $(1+\mu)\big(\frac{\beta}{\mu+\beta}\big)^\beta$. Wage rigidity: $C_{FIXED} = \frac{w_a}{e(w_a, x')}$, $C_{ADJ} = \frac{x'/(1-\beta)}{e_0}$; cutting saves merely 0.2–0.6 pp.

Shapiro–Stiglitz. $\rho V_E = w - e - b(V_E - V_U)$; $\rho V_S = w - (b+q)(V_S - V_U)$; $\rho V_U = a(V_E - V_U)$. NSC: $w = e + \frac{e}{q}\big(\rho + b\frac{L}{L-N}\big)$; labor demand $AF'(eL) = w/e$. $\rho \uparrow$ or $b \uparrow$: NSC up $\Rightarrow N \downarrow, w \uparrow$. $A \uparrow$: labor demand right $\Rightarrow N \uparrow, w \uparrow$. $L \uparrow$: NSC down $\Rightarrow N \uparrow, w \downarrow$.

RBC. $\hat{y}_t = \alpha\hat{k}_t + (1-\alpha)(\hat{A}_t + \hat{l}_t)$; AR(1) shock: $\hat{A}_t = \varepsilon_{A,0}\rho_A^t$ (with $\varepsilon_{A,0} = 0.01$, $\rho_A = 0.95$: 1%, 0.95%, 0.9025%). Rules: $\hat{c} = 0.59\hat{k} + 0.38\hat{A} - 0.13\hat{g}$; $\hat{l} = -0.31\hat{k} + 0.35\hat{A} + 0.15\hat{g}$; $\hat{k}_{t+1} = 0.95\hat{k} + 0.08\hat{A} - 0.004\hat{g}$. Propagation: AR(1) persistence + predetermined capital accumulation. Blanchard–Fischer: Euler $1 - 2\theta E[C_{t+1}] = \frac{1+\rho}{1+A}(1 - 2\theta C_t) \Rightarrow E[C_{t+1}] = C_t$ (random walk); $C_t = AK_t + \frac{A}{1-\phi+A}e_t$, $K_{t+1} = K_t + \frac{1-\phi}{1-\phi+A}e_t$; shock $\varepsilon_t = 1 - \phi + A$: $\Delta C_{t+n} = A$, $\Delta K_{t+n} = 1 - \phi^n$, $\Delta Y_{t+n} = A + \phi^n(1-\phi)$.

Numbers worth memorizing: $\beta = 0.9$, $Y_1 = 100$, $Y_2 = 200$, $r = 10\% \Rightarrow C_1 = 148.33$, $C_2 = 146.84$ (borrower) · $\mu = f = 0.15$, $\beta = 0.06$, $b = 1 \Rightarrow u = 8.1\%$, cost ratio 1.0667; $\beta = 0.03 \Rightarrow 1.0898$ · $u_{EQ} = 6\% \to 9\%$: $\Delta C_{FIXED} = -2.6\%/-1.3\%$, $\Delta C_{ADJ} = -3.2\%/-1.5\%$